Literature Review

Digital Egypt Pioneers Initiative (DEPI) | Data Engineering Final Project

Healthcare Data Pipeline Project

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1. Purpose of This Review

This review compares the main choices for the healthcare data project and explains why the recommended first version uses a small local stack. The goal is not to choose the most advanced platform. The goal is to choose a platform the team can understand, finish, test, and present.

2. Healthcare Data Sources

Two main options are being considered: the existing health and supplement CSV dataset and Synthea synthetic healthcare data. The existing CSV files are useful for early testing because they are already available and easy to inspect. Synthea is another strong option because it provides synthetic patient history, including medications, allergies, encounters, and other healthcare information. The final choice will depend on the project scope, the team's available time, the required dashboard, and how easy the data is to clean and explain.

Source option	Strength	Limitation	Recommendation
Existing health/supplement CSVs	Already available, small, easy to inspect, and good for cleaning practice.	Not a full hospital record and may not represent real hospital operations.	Keep as a working dataset for exploration and testing. The team may use it in the final project if it matches the final requirements.
Synthea	Provides realistic synthetic patient histories and supports formats such as CSV and FHIR. It does not contain real patient data.	More tables and medical concepts to understand.	Keep as much as possible in the final data source or use a small sample if the team decides it is more suitable.
MIMIC-IV or real clinical data	Provides detailed clinical and hospital information for research.	Access rules, training, and privacy requirements make it unsuitable for this beginner project.	Not recommended for the current beginner project.

3. ETL or ELT

ETL means Extract, Transform, Load. The data is cleaned before it reaches the final warehouse tables. ELT means Extract, Load, Transform. The raw data is loaded first, and transformations happen inside the warehouse or lake platform. Both approaches are valid. For this project, ETL is the safer starting choice because the data is small, Python is already familiar to the team, and the rules can be demonstrated step by step. The architecture should keep a Bronze raw layer, so the team can explain how it could move toward ELT later.

Point	ETL	ELT
Best fit	Small or medium-sized projects with clear cleaning rules.	Large cloud platforms with strong SQL or Spark processing.
Learning value	Makes extraction, cleaning, and validation easy to show.	Demonstrates modern cloud warehouse patterns but adds platform concepts.
Project choice	Recommended for version one.	Possible future direction in Fabric or Databricks.

4. Warehouse and SQL Options

SQL Server is a good learning warehouse for the team because it supports familiar T-SQL, relational constraints, views, and Power BI connectivity. Azure SQL Database is a cloud version that can be considered after the local model works. Microsoft Fabric combines data engineering, data integration, warehouse, and Power BI in one SaaS platform, while Fabric Warehouse is designed for relational analytical models such as star schemas. Databricks is powerful for Lakehouse and Spark workloads, but it is more than the team needs for a small CSV project.

Platform	What it is good at	Main concern for this team
SQL Server	Local SQL learning, relational warehouse, clear T-SQL, easy testing.	Someone must install and manage it locally.
Azure SQL	Managed cloud SQL database with less local setup.	Needs an Azure account, network access, and cost.
Microsoft Fabric	Integrated Data Factory, Lakehouse, Warehouse, and Power BI experience.	Many new services and permissions for a beginner project.
Databricks	Large-scale Lakehouse, Spark, notebooks, and advanced engineering.	Too complex for the current data size and team level.

5. Data Model and Dashboard

Microsoft guidance recommends a star schema for Power BI models: dimension tables support filtering and grouping, while fact tables support summarization. The project should therefore avoid joining several fact tables directly. A shared date dimension and user dimension can filter the health and supplement facts safely. The dashboard should be a Power BI report with an overview page, health trends page, supplement and experiment page, and data quality page.

• Overview: total users, health records, average heart rate, average glucose, average sleep, and activity mix.

• Health trends: metrics by date, user age group, and activity level.

• Supplement and experiment: supplement usage, placebo versus non-placebo records, experiment comparison, and dosage.

• Data quality: source row counts, rejected rows, missing values, orphan IDs, and last load batch.

6. Evaluation and Lecturer Feedback

The instructor should assess whether the project is complete, understandable, and supported by evidence. The team should ask for feedback on three checkpoints: after the proposal, after the first working warehouse load, and before the final presentation. Feedback should be recorded as a short action list with the owner and due date. The project should not claim lecturer approval until actual feedback is received.

7. Suggested Improvements

• Add a data dictionary with business definitions and examples for every gold column.

• Keep rejected rows instead of silently deleting them.

• Add batch IDs, load timestamps, source names, and pipeline status to the audit table.

• Add a simple SQL reconciliation report before connecting Power BI.

• Add Synthea only after the current pipeline is stable.

• Use a small number of dashboard pages with clear questions instead of many crowded visuals.

8. Proposed Grading Criteria

Area	Suggested weight	Evidence
Documentation and planning	20%	Proposal, plan, roles, risks, data dictionary, and references
Data sources and modeling	15%	Source explanation, Bronze/Silver/Gold design, and star schema
ETL implementation	25%	Python and SQL scripts, repeatable load, cleaning, and reject handling
Testing and data quality	15%	Row counts, reconciliation, rule checks, and test results
Power BI dashboard	15%	Useful measures, relationships, layout, and clear insights
Presentation and teamwork	10%	Clear explanation, contribution from each member, and response to questions

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References Used

- MITRE Synthea official site: https://synthetichealth.github.io/synthea/
- MITRE Synthea downloads and formats: https://synthetichealth.github.io/downloads.html
- Microsoft Fabric overview: https://learn.microsoft.com/en-us/fabric/fundamentals/microsoft-fabric-overview
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